

SFO Through Crisis: A Passenger Story

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Abstract—San Francisco International Airport (SFO) is one of the busiest airports in the United States, consistently praised for its amenities and passenger experience. Behind that reputation, however, is a traffic story shaped by forces entirely outside the airport’s control. This project investigates how SFO has performed during large-scale disruptions, specifically 9/11, the 2008 recession, and COVID-19, by exploring over 25 years of monthly enplaned passenger data. The central questions are how deep each crisis cut into passenger volume and what recovery looked like in the aftermath. Using an interactive scrollytelling visualization built with D3.js and Scrollama.js, four coordinated views (a calendar heat map, small multiples pie charts, small multiples line charts, and a stream graph) guide users through seasonal traffic patterns, shifts in international destination composition, domestic versus international recovery rates, and structural changes in the airline market at SFO. The system was designed to make these patterns accessible to a wide range of audiences, regardless of prior experience with data visualization. Across all three crises, SFO recovered, but the depth and duration varied significantly, with COVID-19 standing out as the most severe contraction in the airport’s recorded history by a wide margin.

I. INTRODUCTION

SFO is consistently ranked among the best airports in the United States, known for its museum exhibits, sensory rooms, and quality dining. But behind that reputation is a passenger traffic history shaped by forces entirely outside the airport’s control. Major disruptions like 9/11, the 2008 recession, and COVID-19 each left a measurable mark on SFO’s monthly passenger volumes, and understanding what those marks look like in the data, and how recovery unfolded each time, is the central motivation for this project.

Airport traffic data is not often explored through the lens of crisis and recovery. Most passengers experience disruptions as news headlines or canceled flights, not as patterns in a dataset. By reading this paper, you will come away with a clear, comparative picture of how severe each disruption was at SFO, how long recovery took each time, and whether different segments of air travel – domestic versus international, or individual carriers – were affected differently. That kind of comparative insight is difficult to extract from news coverage alone, and is relevant to airport planners, policy researchers, and anyone interested in how large infrastructure systems respond to external shocks.

The dataset, published by the SF Airport Commission through DataSF [13], covers monthly traffic records from July 1999 to the present across airlines, terminals, and geographic

regions. That range captures not just the drops, but the full recovery arcs, making it possible to compare how deep each crisis cut relative to the others. The interface is a scrollytelling site built with D3.js, where each scroll step reveals a new visualization with a short narrative caption. Rather than presenting everything at once, the format guides the user through the data progressively, moving from a broad 25-year overview down to individual carrier trends.

A. Project Objectives

- **Objective 1:** Reveal seasonal travel patterns at SFO and show how those patterns were disrupted during each major crisis.
- **Objective 2:** Visualize how SFO’s international passenger mix was distributed across destination regions, and how that composition shifted across eras.
- **Objective 3:** Compare whether domestic and international routes recovered at the same pace after each disruption.
- **Objective 4:** Track how the airline market at SFO shifted over time, including which carriers grew, which pulled back, and which exited entirely.

II. RELATED WORK

A. Narrative Visualization and Scrollytelling

Segel and Heer [1] laid the groundwork for thinking about data visualization as storytelling, introducing a spectrum between fully author-driven narratives and fully reader-driven exploration. Scrollytelling sits in a hybrid space: the author controls pacing and sequence, but the reader can still hover, isolate, and explore within each step. That structure fit this project well because each crisis has a natural before, during, and after that benefits from pacing rather than dumping everything at once. Satyanarayan and Heer’s Lyra [2] showed how layered interactive systems can carry a temporal narrative effectively. For implementation, The Pudding’s scrollytelling tutorial [3] and Shorthand’s catalog of real-world examples [4] were both useful reference points. The Scrollama library [35], which handles scroll-triggered step detection, was used directly in this project.

B. Aviation and Crisis Recovery Research

Sun et al. [5] conducted a data-driven analysis of aviation recovery from COVID-19, including findings on differential recovery rates between domestic and international routes.

Those findings map closely to what shows up in the SFO data. Mumbower [6] specifically studied airline market exit behavior following COVID-19, which informed the stream graph’s focus on which carriers shrank or left SFO entirely. The Airports Council International [7] report on COVID-19’s impact on global airports provided industry-wide context for benchmarking SFO’s recovery against broader trends.

C. Time Series and Temporal Visualization

Van Wijk and Van Selow [8] established the calendar heat map as a technique for revealing periodicity and anomalies in long time series, with months on one axis, years on the other, and color encoding magnitude. That design was adopted directly for the first scroll step, and Bostock’s Observable calendar example [31] served as a concrete D3 reference for the grid layout and color scale. Havre et al.’s ThemeRiver [9] introduced the stream graph as a way to track how categorical compositions shift over time, which is exactly what the airline market visualization needed. Bostock’s stream graph example [29] provided a starting point for the D3 stack and area generator configuration. All four views were built on D3: Data-Driven Documents [10], which was the primary technical foundation throughout.

D. Visualization Design and Chart-Specific References

Munzner’s *Visualization Analysis and Design* [11] shaped several encoding decisions, especially the choice to use position on a shared axis for the line chart rather than color or area, which is the more accurate channel for comparing magnitude. Cairo’s *The Functional Art* [12] reinforced the idea that chart type should serve the task, not the other way around. That came up directly when user testing showed the chord diagram was too hard to read for a general audience, and the view was redesigned entirely.

The small multiples pie chart layout was informed by Bostock’s small multiples example [32], which showed how to coordinate scales and encodings across repeated chart instances. The chord diagram prototype was built using the D3 chord diagram example [30] and the D3 Graph Gallery chord reference [27] before being replaced. The synchronized crosshair on the line chart was adapted from Bostock’s line chart with tooltip example [33], which demonstrates bisect-based hover snapping across a time series. The D3 Graph Gallery heatmap [24], line chart [28], and stream graph [25] references were consulted throughout for layout and axis configuration.

III. APPROACH

A. Data Source and Processing

The dataset is the Air Traffic Passenger Statistics file from the SF Airport Commission via DataSF [13]. It contains monthly records starting in July 1999, updated quarterly. The key columns for this project were `Activity_Period`, `Operating_Airline`, `GEO_Summary` (Domestic or International), `GEO_Region` (US, Asia, Europe, Canada, Mexico, etc.), `Activity_Type_Code` (Enplaned, Deplaned, or Thru/Transit), and `Passenger_Count`.

All processing was done in-browser with D3.js. A few things needed cleaning. `Activity_Period` stores dates as six-digit integers (202012 means December 2020), so those were parsed into JavaScript Date objects for time-series work. Everything was filtered to `Activity_Type_Code = "Enplaned"` before aggregating, because an airline can appear across multiple rows per month for different terminals and regions, making it easy to accidentally double-count. Airline names also needed standardizing to account for rebranding and mergers across 25 years. `GEO_Region` labels were grouped into a consistent set of destination categories for the pie charts. The final visualizations run from January 2000 to December 2024 to avoid partial-year edge effects at both ends of the dataset.

B. Scrollytelling Structure

The site is a single-page scrollytelling application built with HTML, CSS, and D3.js, using the Scrollama library to detect scroll progress and trigger transitions at each step. The four scroll steps map directly to the four objectives: calendar heat map (Objective 1), small multiples pie charts (Objective 2), small multiples line chart (Objective 3), and stream graph (Objective 4). Each visualization is accompanied by a narrative caption that provides historical context. Crisis bands and event annotations are applied consistently across all time-based views so users can cross-reference the same events across different perspectives.

Executing this approach required several moving parts. Scrollama handled detecting when the user scrolled to each step and triggered the corresponding chart to appear. Each of the four visualizations was built separately with its own data filtering, sizing, and hover behavior, so they would not interfere with each other. Specific interactions included row and column highlighting on the heat map, synchronized hover across all six pie charts, a crosshair that tracked both line chart panels at once, and airline highlighting on the stream graph. Crisis bands were built as a shared component applied to every time-based chart so the same historical windows line up visually across all views.

C. Design Iterations and Problems Encountered

The biggest change mid-project was replacing the chord diagram originally planned for Objective 2. The chord diagram was meant to show passenger flows between SFO and destination regions, but during informal user testing, participants had real trouble reading it, especially when trying to compare distributions across different time periods. The small multiples pie chart layout turned out to be much more readable. It also made it easier to support direct comparison across the six eras: pre-9/11, post-9/11, pre-recession, recession, pre-COVID, and COVID/recovery.

A log scale was also explored for the calendar heat map. It did reveal just how extreme COVID-19’s impact was relative to everything before it, which was striking. But it compressed the 2008 recession dip enough that the recession became hard to notice, even though it was a real and significant disruption.

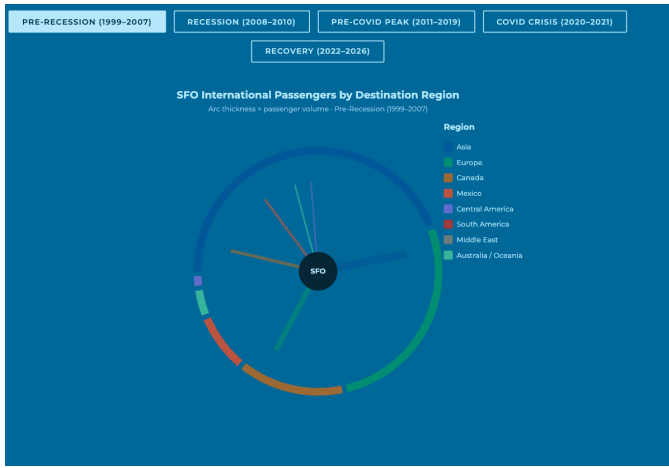


Fig. 1. Early chord diagram prototype for Objective 2. Users had difficulty extracting meaningful comparisons across time periods, which prompted the switch to small multiples pie charts.



Fig. 2. Calendar heat map using a logarithmic color scale. While the log scale makes COVID-19's impact dramatically more visible relative to prior years, it compresses the 2008 recession dip to the point where it becomes difficult to distinguish from normal seasonal variation. The linear scale was kept for the final design to preserve the visibility of all three crises as distinct events.

The project's narrative depends on all three crises being visible as distinct events, so the linear scale was kept.

Other changes were easier to address. Charts were made larger after feedback that they were hard to read on laptop screens. Headings and subheadings were added above each visualization. More narrative text was added throughout. The dataset was trimmed to January 2000 through December 2024 so all four views share consistent time boundaries, as the raw data began partway through 1999 and partway through 2026.

IV. RESULTS

A. Calendar Heat Map (Objective 1)

The heat map shows monthly total enplaned passenger counts from 2000 to 2024, with months on the x-axis and years

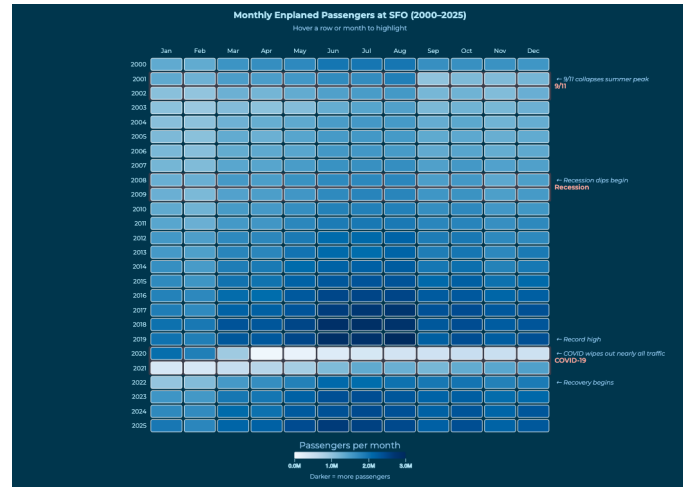


Fig. 3. Calendar heat map of monthly SFO enplaned passengers from 2000 to 2024. Darker cells indicate higher passenger volumes. The 2020–2021 washout reflects COVID-19. Summer peaks are visible in every non-crisis year. Hovering the edge of a row or column isolates that year or month across the full chart.

on the y-axis. A sequential single-hue color scale encodes volume; darker means more passengers. Three patterns come through immediately: summer peaks in July and August every year, a clear trough after 9/11 in late 2001, a subtler but visible dip during the 2008–2009 recession, and then 2020 as a near-complete washout that carries into 2021.

Hovering the left or right edge of a row dims all other years to 15% opacity for year-level focus. Hovering the top of a column does the same for a single month. A tooltip shows the exact month, year, and passenger count. Annotations on the right side of the chart label each disruption event, so users know what to look for and can follow along without any prior knowledge of SFO's history.

B. Small Multiples Pie Charts (Objective 2)

Six pie charts represent six distinct eras: pre-9/11 (2000–2001), post-9/11 (2002–2003), pre-recession (2004–2007), recession (2008–2009), pre-COVID (2010–2019), and COVID/recovery (2020–2024). Each chart shows the share of international enplaned passengers by destination region, with consistent color encoding across all six.

Asia's share of SFO's international traffic contracted sharply during COVID-19, and the recovery period shows some regions bouncing back faster than others. Hovering any slice or legend item fades all other regions to 20% opacity across all six charts at once, making it easy to track one region's trajectory from era to era without losing the overall composition.

C. Small Multiples Line Chart (Objective 3)

Two side-by-side line charts share the same y-axis scale, one for domestic and one for international monthly enplaned passengers. Shaded crisis bands are aligned across both panels for the 9/11 window, the 2008–2009 recession, and COVID-19. A crosshair hover synced across both panels snaps to



Fig. 4. Small multiples pie charts showing SFO international passenger distribution by destination region across six historical eras. Hovering a legend item or pie slice highlights that region across all six charts simultaneously. Tooltips show region name, total passenger count, and share of that era's international total.

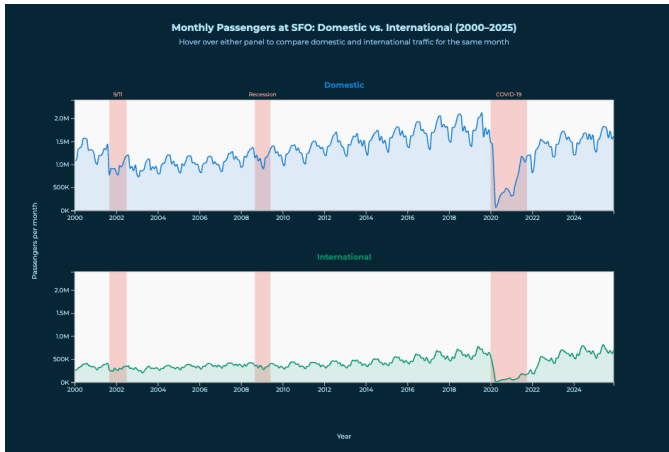


Fig. 5. Side-by-side line charts comparing monthly domestic (left) and international (right) enplaned passengers at SFO from 2000 to 2024. Panels share the same y-axis scale. A synchronized crosshair lets users compare both segments for any given month.

the nearest data point and shows domestic and international passenger counts for that month.

Domestic traffic bounced back faster after both the 2008 recession and COVID-19. Following COVID-19, domestic volumes were approaching pre-pandemic levels by late 2022, while international volumes were still below 2019 levels through the end of the dataset. The shared y-axis makes that gap visible without any annotation needed.

D. Stream Graph (Objective 4)

The stream graph shows the top airlines at SFO over time, with each band's width proportional to that airline's annual share of enplaned passenger volume. Crisis bands mark each disruption. Hovering over a band shows the year and annual passenger count. Hovering a legend item highlights that airline's stream and fades all others.

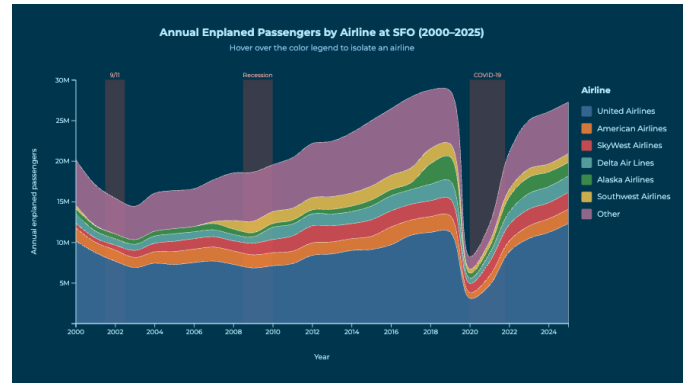


Fig. 6. Stream graph showing annual enplaned passenger share for the top airlines at SFO from 2000 to 2024. Band width reflects each airline's share of total volume. Crisis bands mark each disruption. Hovering a legend item isolates that airline's stream across the full time range.

The stream graph shows patterns that aggregate passenger counts cannot reveal. United Airlines dominated SFO for the entire 25-year period, reflecting its role as a hub carrier with structural advantages other airlines cannot easily replicate. After 9/11, several smaller carriers pulled back capacity and never fully returned. The 2008 recession caused a broad contraction across all carriers, but the relative order held. COVID-19 was a different kind of disruption. Total volume collapsed in 2020, and the 'Other' category shrank noticeably during recovery, suggesting smaller carriers exited SFO at a higher rate than major carriers did. By 2024, total volume is approaching pre-COVID levels, but the airline composition looks meaningfully different than it did in 2019.

E. Measuring Success

Success was measured through informal usability sessions with friends and classmates who had varying levels of visualization familiarity, ranging from non-technical users to CS students. The main criterion was whether someone with no prior knowledge of SFO's history could identify all three major disruptions and say something meaningful about at least one of them after a single scroll-through. Most participants were able to do this after the final round of revisions. Earlier sessions surfaced recurring pain points: the chord diagram was difficult to interpret, charts were too small to read comfortably, and without narrative captions or headings it was difficult to know what each visualization was showing or how to use it. Those observations drove the most significant changes: replacing the chord diagram with small multiples pie charts, enlarging the charts, and adding headings and narrative captions above each visualization.

V. DISCUSSION

The scrollytelling approach was a good fit for this project. Because each crisis has a clear before, during, and after, the scroll-driven narrative makes those moments feel significant rather than just being a dip on a chart or a lighter-colored cell in a grid. Starting with the heat map gives users the full 25-year picture first, so by the time they reach the line chart or

stream graph, the crisis windows are already familiar reference points.

Replacing the chord diagram was the right call, even if it was frustrating at the time. Chord diagrams are expressive, but they ask a lot of the reader, especially someone who has not seen one before. The small multiples layout is simpler and lets the comparison do the work without requiring visualization literacy to decode the chart type first. That experience reinforced something from Munzner [11] and Cairo [12]: expressive power only matters if the audience can actually read the chart.

The log scale experiment produced an unexpected tradeoff. It made it clear just how much COVID-19 dwarfed every previous disruption; the 2008 recession barely registers on a log scale relative to 2020. But that same compression made the recession feel negligible, which is not accurate. The linear scale keeps all three crises legible as distinct events, which the narrative depends on.

This project was my first time building a scrollytelling site from scratch, and working with Scrollama and D3 together taught me a lot about how to structure an interactive visualization pipeline. Getting scroll-triggered transitions to work reliably across four different chart types required understanding how D3 manages state and selections in a way that reading documentation alone did not fully convey. On the design side, the project clarified how much audience awareness shapes every decision. Watching someone not know where to start reading the chord diagram was more instructive than any design reading I did this semester, and it pushed me to prioritize comprehension over expressiveness for every chart choice that followed.

The biggest thing I would have done differently is start user testing earlier. The chord diagram problem was caught during informal walkthroughs, but by that point the implementation was already done. Earlier paper prototype testing, even just sketches, would have surfaced that issue before any code was written. The fix was worth making, but the timing made it stressful.

VI. FUTURE WORK

If more time were available, three additions would have the most impact. First, a grid zoom animation during scroll transitions would bridge the calendar heat map's broad overview into the detail views that follow, making the shift in scale feel intentional rather than abrupt.

Second, adding source-cited written annotations directly inside each visualization, drawn from journalism and research, would give the narrative more credibility and depth. Right now the context lives in the captions; surfacing specific statistics or quotes closer to the data they describe would strengthen the storytelling.

Third, hover-to-highlight interactions currently reset when the cursor moves away. A click-to-stay mode would let users lock a selected region, airline, or time window in place while continuing to scroll.

A longer-term extension would be adding other West Coast airports (LAX, SEA) to let users compare SFO's recovery

trajectory against comparable hubs and see whether its patterns are typical for the region or specific to its route mix.

VII. CONCLUSION

Twenty-five years of SFO passenger data tells a clear story: the airport has weathered every major disruption it has faced, but each recovery looked different. The scrollytelling system built for this project makes that story accessible to someone with no background in visualization or aviation, moving from a broad calendar overview down to individual airline trends across four scroll steps. The most useful design lesson was that chart type needs to serve the audience, not just the data. The switch from chord diagram to small multiples was a direct result of watching real users struggle with the original design. More broadly, this project suggests that scrollytelling is a viable format for making long-term infrastructure data accessible to a general audience, one that could extend to other airports, transit systems, or any domain where crisis and recovery play out over years rather than days. Future work will focus on richer annotations, persistent interactions, and cross-airport comparison.

REFERENCES

- [1] E. Segel and J. Heer, "Narrative Visualization: Telling Stories with Data," *IEEE Transactions on Visualization and Computer Graphics*, vol. 16, no. 6, pp. 1139–1148, Nov. 2010. <https://doi.org/10.1109/tvcg.2010.179>
- [2] A. Satyanarayan and J. Heer, "Lyra: An Interactive Visualization Design Environment," *Computer Graphics Forum*, vol. 33, no. 3, pp. 351–360, 2014. <https://doi.org/10.1111/cgf.12391>
- [3] The Pudding, "How to Implement Scrollytelling with Six Different Libraries," 2017. [Online]. Available: <https://pudding.cool/process/how-to-implement-scrollytelling/>
- [4] K. Abbamonte, "15 Engaging Scrollytelling Examples to Inspire Your Content," Shorthand.com. [Online]. Available: <https://shorthand.com/the-craft/scrollytelling-examples/index.html>
- [5] X. Sun et al., "A Data-Driven Analysis of the Aviation Recovery from the COVID-19 Pandemic," *Journal of Air Transport Management*, vol. 109, p. 102401, June 2023. <https://doi.org/10.1016/j.jairtraman.2023.102401>
- [6] S. Mumbower, "Airline Market Exit after a Shock Event: Insights from the COVID-19 Pandemic," *Transportation Research Interdisciplinary Perspectives*, vol. 14, p. 100621, June 2022. <https://doi.org/10.1016/j.trip.2022.100621>
- [7] Airports Council International, "The Impact of COVID-19 on Airports and the Path to Recovery," Feb. 2023. [Online]. Available: <https://aci.aero/2023/02/22/the-impact-of-covid-19-on-airportsand-the-path-to-recovery-industry-outlook-for-2023/>
- [8] J. J. van Wijk and E. R. van Selow, "Cluster and Calendar Based Visualization of Time Series Data," in *Proc. IEEE Symposium on Information Visualization (InfoVis)*, 1999, pp. 4–9.
- [9] S. Havre, E. Hetzler, P. Whitney, and L. Nowell, "ThemeRiver: Visualizing Thematic Changes in Large Document Collections," *IEEE Transactions on Visualization and Computer Graphics*, vol. 8, no. 1, pp. 9–20, 2002.
- [10] M. Bostock, V. Ogievetsky, and J. Heer, "D3: Data-Driven Documents," *IEEE Transactions on Visualization and Computer Graphics*, vol. 17, no. 12, pp. 2301–2309, 2011.
- [11] T. Munzner, *Visualization Analysis and Design*. Boca Raton, FL: CRC Press, 2014.
- [12] A. Cairo, *The Functional Art: An Introduction to Information Graphics and Visualization*. Berkeley, CA: New Riders, 2012.
- [13] SF Airport Commission, "Air Traffic Passenger Statistics," DataSF Open Data Portal. [Online]. Available: <https://data.sfgov.org/Transportation/Air-Traffic-Passenger-Statistics/rkru-6vcg>
- [14] M. Bostock, "D3.js: Data-Driven Documents," [Online]. Available: <https://d3js.org/>

- [15] M. Bostock, “d3-scale,” [Online]. Available: <https://d3js.org/d3-scale>
- [16] M. Bostock, “d3-scale-chromatic,” [Online]. Available: <https://d3js.org/d3-scale-chromatic>
- [17] M. Bostock, “d3-axis,” [Online]. Available: <https://d3js.org/d3-axis>
- [18] M. Bostock, “d3-shape,” [Online]. Available: <https://d3js.org/d3-shape>
- [19] M. Bostock, “d3-selection,” [Online]. Available: <https://d3js.org/d3-selection>
- [20] M. Bostock, “d3-chord,” [Online]. Available: <https://d3js.org/d3-chord>
- [21] M. Bostock, “d3-time-format,” [Online]. Available: <https://d3js.org/d3-time-format>
- [22] M. Bostock, “d3-array,” [Online]. Available: <https://d3js.org/d3-array>
- [23] M. Bostock, “d3-transition,” [Online]. Available: <https://d3js.org/d3-transition>
- [24] The D3 Graph Gallery, “Heatmap,” [Online]. Available: <https://d3-graph-gallery.com/heatmap.html>
- [25] The D3 Graph Gallery, “Stream Graph,” [Online]. Available: <https://d3-graph-gallery.com/streamgraph.html>
- [26] The D3 Graph Gallery, “Area Chart,” [Online]. Available: <https://d3-graph-gallery.com/area.html>
- [27] The D3 Graph Gallery, “Chord Diagram,” [Online]. Available: <https://d3-graph-gallery.com/chord.html>
- [28] The D3 Graph Gallery, “Line Chart,” [Online]. Available: <https://d3-graph-gallery.com/line.html>
- [29] M. Bostock, “Streamgraph,” Observable. [Online]. Available: <https://observablehq.com/@d3/streamgraph>
- [30] M. Bostock, “Chord Diagram,” Observable. [Online]. Available: <https://observablehq.com/@d3/chord-diagram>
- [31] M. Bostock, “Calendar,” Observable. [Online]. Available: <https://observablehq.com/@d3/calendar>
- [32] M. Bostock, “Small Multiples,” Observable. [Online]. Available: <https://observablehq.com/@d3/small-multiples>
- [33] M. Bostock, “Line Chart with Tooltip,” Observable. [Online]. Available: <https://observablehq.com/@d3/line-chart-with-tooltip>
- [34] M. Bostock, “Stacked Area Chart,” Observable. [Online]. Available: <https://observablehq.com/@d3/stacked-area-chart>
- [35] R. Samora, “Scrollama.js,” GitHub. [Online]. Available: <https://github.com/russellsamora/scrollama>
- [36] Mozilla Developer Network, “Intersection Observer API,” [Online]. Available: https://developer.mozilla.org/en-US/docs/Web/API/Intersection_Observer_API
- [37] Mozilla Developer Network, “CSS color-mix(),” [Online]. Available: https://developer.mozilla.org/en-US/docs/Web/CSS/color_value/color-mix
- [38] Mozilla Developer Network, “CSS clamp(),” [Online]. Available: <https://developer.mozilla.org/en-US/docs/Web/CSS/clamp>
- [39] Mozilla Developer Network, “SVG clipPath,” [Online]. Available: <https://developer.mozilla.org/en-US/docs/Web/SVG/Element/clipPath>
- [40] Google, “Google Fonts,” [Online]. Available: <https://fonts.google.com/>
- [41] Anthropic, “Claude AI,” used for data cleaning assistance (parsing Activity_Period, filtering to Enplaned rows, standardizing airline names) and responsive layout support. [Online]. Available: <https://claude.ai>
- [42] Professor Joshi, “Scrollama Examples,” provided course materials, University of San Francisco, CS 360, Spring 2026.